

Deployment Guide

Configuration, local Docker, Railway and how to start a new project from the template.

Project	Digital Twin Catalog (NARAGA 3D Catalog), reusable 3DCityDB template
Document	Deployment
Version	0.1 draft for review, 4 October 2026
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Introduction

Status: **draft v0.1 for review**. Items marked *verify* are assumptions to check before relying on them.

Configuration (env vars)

Var	Meaning
<code>DATABASE_URL</code> / <code>POSTGRES_*</code>	DB connection (API and migrations)
<code>CITYDB_SRID</code> , <code>CITYDB_HEIGHT_EPSG</code>	Fixed at the first start of the db volume (e.g. 32750 / 4979)
<code>SECRET_KEY</code> , <code>DEBUG</code> , <code>ALLOWED_HOSTS</code>	Django
<code>ALLOWED_ORIGINS</code>	CORS allow-list for the frontend
<code>PROJECT</code>	Seed folder name under <code>db/seeds/</code>

`.env.example` is committed; `.env` is git-ignored.

Local

`make up (db + api) → make migrate → make seed → open http://localhost:8000/admin, /viewer/, /api/v1/docs . make test` runs migration + invariant tests. Pinned image: `3dcitydb/3dcitydb-pg:16-3.4-5.1.4`; `citydb-tool 1.4.0`.

Railway

Services (same Dockerfiles as local, `railway.toml` per service): 1. **db** - image `3dcitydb/3dcitydb-pg:16-3.4-5.1.4`, volume mounted at the Postgres data dir, private networking only, env `SRID`, `HEIGHT_EPSG`, `POSTGRES_PASSWORD`. *verify*: init works on a Railway volume (may need `PGDATA` subdirectory), volume size/plan cost for the LOD1 data, image runs on Railway's amd64 hosts. 2. **api** - Django; pre-deploy command `dbmate up && psql -f db/seeds/$PROJECT/*.sql`; healthcheck `/api/v1/health`; public domain; `DATABASE_URL` from the private network. 3. **backup** (cron) - nightly `pg_dump -Fc` to an S3-compatible bucket. Do not rely on volume snapshots alone. 4. `citydb-tool imports` run as a one-off container (local or Railway) over the private network or a temporary TCP proxy; close the proxy afterwards.

Steps: create project → add db service (volume) → wait healthy → add api service from the GitHub repo → set env → deploy → create first admin user (`python manage.py createsuperuser` via Railway shell) → run `newman` smoke test against the Railway environment. Rollback: redeploy the previous api deployment; DB changes are forward-only migrations (use `dbmate down` only for the latest, before data depends on it); data: restore from dump.

New project from the template

Copy/fork the repo → change `.env` (SRID, project name) → add `db/seeds/<project>/` → make up migrate seed. Add themes by inserting into `ref_theme` (admin UI).